

MatEnsemble: A Flux-native Python workflow manager for adaptive high-throughput computing on HPC systems

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Summary

MatEnsemble is a Python package for defining and running high-throughput workflows on High-Performance Computing (HPC) systems. Users construct a Directed Acyclic Graph (DAG) of *chores*, where each chore is either a delayed python function call or an executable command, each with explicit resource requirements. MatEnsemble submits these chores through the Flux resource manager, tracks completion, resolves dependencies through serialized results, and records workflow state in a structured layout. The package is designed for...

where launching one batch job per task would create excessive scheduler overhead or leave resources idle.

Statement of need

Modern materials science workloads increasingly consist of large ensembles of related simulations rather than a single monolithic calculation. Examples include

These workloads may require thousands or even millions of relatively small tasks whose execution patterns are difficult to express efficiently using traditional batch schedulers.

Submitting large numbers of short-lived jobs directly to a system scheduler such as SLURM can create significant scheduling overhead, increase queue wait times, and place unnecessary load on shared scheduling infrastructure. For this reason some systems restrict the number of invocations of certain commands to reduce the strain that could be placed on the scheduler. On the other hand, grouping work into batches within a single allocation often leads to poor utilization, as fast-running tasks complete early and leave resources idle while longer-running tasks continue to execute. Researchers need a mechanism that can efficiently manage large collections of heterogeneous tasks while maintaining high utilization of expensive HPC resources.

MatEnsemble addresses these challenges by combining a Python-native workflow interface with the Flux resource manager. Rather than submitting thousands of independent scheduler jobs, users acquire a single allocation and allow MatEnsemble to manage task execution within a user-space Flux instance. This hierarchical scheduling model dramatically reduces scheduler overhead while enabling fine-grained control over task placement and execution. MatEnsemble continuously monitors available resources and adaptively backfills newly eligible tasks as running work completes, maintaining high utilization even when task runtimes vary significantly.

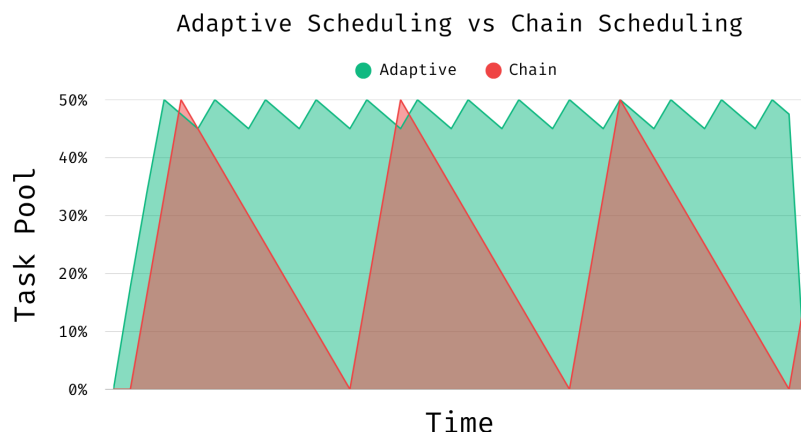


Figure 1: Adaptive Workflow Management

38 Workflows may consist of Python functions, MPI applications, shell commands, or combinations
39 thereof. For example, a user may define an MPI-enabled chore that launches multiple ranks
40 through Flux while still participating in the same dependency-aware workflow:

```
from matensemble.pipeline import Pipeline
from mpi4py import MPI

pipe = Pipeline()

@pipe.chore(num_tasks=10, cores_per_task=1, mpi=True)
def mpi_helloworld():
    size = MPI.COMM_WORLD.Get_size()
    rank = MPI.COMM_WORLD.Get_rank()
    name = MPI.Get_processor_name()

    print(f"Hello World! I am process {rank} of {size} on {name}.")

for _ in range(10):
    mpi_helloworld()

pipe.submit()
```

41 Each invocation becomes an independent Flux job while remaining under the control of the
42 workflow manager and sharing the same allocation.

43 Unlike many existing scientific workflow systems that rely on external databases, centralized
44 services, or privileged scheduler interactions, MatEnsemble is designed as a lightweight Python
45 library that operates entirely within the user's allocation. Workflows are expressed as DAG
46 of Python callables or executable tasks, with declarative resource requirements attached to
47 each task. The framework leverages Flux's scalable user-space scheduling architecture while
48 exposing a familiar Python programming model for workflow construction and execution.

49 A distinguishing feature of MatEnsemble is its support for user-defined scheduling strategies. In
50 addition to the built-in adaptive scheduler, users may inject custom workflow logic capable of
51 dynamically generating new tasks during execution based on intermediate results. This enables

52 the construction of dynamically expanding scientific workflows and active-learning loops.

53 The target users are computational scientists and HPC developers who need to compose
54 ensembles of Python functions, MPI programs, shell commands, and analysis steps without
55 writing a custom scheduler. The package also targets research software developers who want a
56 lightweight execution layer for Flux-enabled systems while retaining human readable files for
57 debugging and reproducibility.

58 State of the field

59 Several mature Python workflow systems already support scientific task graphs. Parsl provides
60 a broad parallel scripting model for Python functions and external applications across local,
61 cluster, cloud, and grid resources. Jobflow provides a Pythonic decorator-based workflow
62 model aimed at high-throughput computational workflows, with strong adoption in materials
63 science. libEnsemble focuses on dynamic ensembles using a generator-simulator-allocator model,
64 particularly for adaptive sampling and optimization campaigns. These systems demonstrate the
65 value of Python-native workflows for computational science, and MatEnsemble is complementary
66 rather than a replacement.

67 Example workflow

68 The primary user interface in MatEnsemble is the Pipeline object. Python functions decorated
69 with `@matensemble.pipeline.Pipeline.chore()` become delayed function calls that return
70 `OutputReference` objects rather than executing immediately. These references can be passed
71 into later chores, allowing MatEnsemble to infer dependencies automatically and construct a
72 DAG. The example below computes the factorial of 100 and then computes the digit sum of the
73 resulting value. The dependency between the two chores is inferred from the `OutputReference`
74 returned by the first call.

```
from matensemble.pipeline import Pipeline

pipe = Pipeline()

@pipe.chore()
def factorial(n):
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result

@pipe.chore()
def digit_sum(n):
    return sum(int(x) for x in str(n))

a = factorial(100)
b = digit_sum(a)

pipe.submit()
```

75 Internally, MatEnsemble transforms workflows such as this into a DAG of chores, validates
76 dependencies, serializes the workflow state, and executes eligible tasks through Flux as resources
77 become available.

78 Software design

79 MatEnsemble separates workflow definition, scheduling, and execution into distinct components.
80 Users construct workflows through the Pipeline API, which records delayed chores and

81 dependency relationships. During submission, the workflow is validated, serialized, and handed
82 to a FluxManager instance that coordinates execution through Flux.

Workflow Lifecycle

Figure 2: Workflow Lifecycle

83 MatEnsemble's architecture is centered on a separation between workflow definition, scheduling,
84 and execution. The user process constructs a graph of delayed chores through the Pipeline
85 API, while FluxManager owns runtime state such as blocked, ready, running, completed, and
86 failed chore sets. Individual chores are submitted through Flux as independent jobs. For Python
87 chores, this separation creates a reconstruction problem. Callables defined in the user's Python
88 process are not available by memory reference inside a fresh worker process launched by Flux.

89 MatEnsemble solves this by creating a registry where functions are serialized and stored by
90 value. Each call to one of these registered functions then creates a specification that is placed
91 into the output directory of the respective chore. MatEnsemble is then able to submit an
92 internal module, `matensemble.runtime_worker`, as a Flux job which will reload the chore,
93 resolve dependency outputs, execute the callable, and write a result artifact for downstream
94 chores.

95 During workflow construction, `matensemble.pipeline.Pipeline.chore` records Python
96 function calls and `matensemble.pipeline.Pipeline.exec` records executable commands.
97 When one chore uses the output of another, MatEnsemble automatically detects this
98 relationship and creates a dependency link between the two chores. Before execution begins,
99 MatEnsemble builds a DAG representing the workflow, verifies that all dependencies are valid,
100 and topologically sorts the graph to determine the correct execution order.

101 At runtime, MatEnsemble creates a dedicated directory for each chore to store outputs, logs,
102 metadata, and results. The chore object itself is also serialized to allow the arguments to
103 a function call be any python object rather than restricting it to JSON serializable objects.
104 When the workflow is launched, a workflow directory is created that contains all of the files
105 needed to execute, monitor, and debug the workflow.

```
106 <basedir or cwd>/
107   └─ matensemble_workflow-YYYYMMDD_HHMMSS/
108       ├── status.json           # Atomically updated for the dashboard / monitoring
109       ├── matensemble_workflow.log # Detailed text log from the logger
110       └─ out/
111           ├── registry/         # Pickled chore callables
112               ├── func_qualname_1
113               ├── ...
114               └─ func_qualname_n
115           ├── <chore_id_1>/
116               ├── stdout
117               ├── stderr
118               ├── metadata.json  # Metadata of the chore in JSON for debugging
119               ├── chore.pickle   # Pickled chore object
120               └─ result.pickle   # Python chore return value
121           ├── ...
122           └─ <chore_id_n>/
123               └─ ...
```

124 At runtime, FluxManager owns the ready, blocked, running, completed, and failed chore sets.
125 The manager continuously monitors resource availability, submits ready chores that fit within
126 the allocation, processes completed tasks, updates dependency information, and unblocks

127 newly eligible work. This scheduling loop continues until no ready, running, or blocked chores
128 remain.

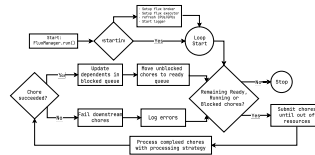


Figure 3: Manager Loop

129 The scheduler uses the *strategy pattern* to separate completion handling logic from the core
130 scheduling loop. A strategy determines how completed chores are processed and how newly
131 available work is admitted into the workflow.

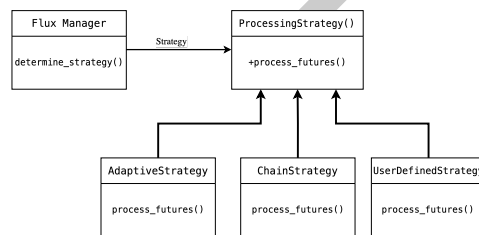


Figure 4: Strategy Pattern

132 The default adaptive strategy attempts to submit newly unblocked chores immediately as
133 resources become available, helping maintain high utilization within a running allocation. The
134 non-adaptive strategy provides a simpler wave-based execution model in which newly eligible
135 chores are submitted during the next scheduling cycle.

136 MatEnsemble also supports user-defined strategies. These strategies can inspect the results of
137 completed Python chores and dynamically generate additional work during execution. This
138 capability enables adaptive parameter searches, active-learning workflows, iterative optimization
139 campaigns, and other data-driven scientific applications where the total amount of work is not
140 known in advance.

141 Dynamic workflow expansion

142 User-defined strategies allow workflows to expand dynamically during runtime. A strategy
143 may inspect the result of a completed chore and return a `matensemble.chore.ChoreSpec`
144 describing new work to be added to the workflow. The new chore is validated, inserted into
145 the dependency graph, and scheduled like any other task.

146 The example below implements a simple binary search. Each completed guess chore produces
147 a result that is examined by a user-defined strategy. To attach chores to a strategy by adding
148 their name to the BOLO List. This will tell the manager to Be On the Look-Out (BOLO) for
149 this chore, if it sees it then it will spawn the strategy and pass the results from the chore to
150 the strategy. Based on that result, the strategy generates another guess chore until the target
151 value is found.

```
@pipe.chore()
def guess(lower: int, upper: int, guess_num: int = 1) -> dict:
    return {
        "guess": ((lower + upper) // 2),
        "low": lower,
        "high": upper,
```

```

        "num_guesses": guess_num,
    }

    answer = random.randint(1, 100)
    bolo_list = ["guess"]

    @pipe.strategy(bolo_list=bolo_list)
    def higher_or_lower(guess_result, ans=answer):

        if guess_result["guess"] == ans:
            print(
                f"Good Job! I was thinking of {ans}, "
                f"and you got it in {guess_result['num_guesses']}"
            )

        elif guess_result["guess"] < ans:
            return ChoreSpec(
                args=(
                    guess_result["guess"] + 1,
                    guess_result["high"],
                    guess_result["num_guesses"] + 1,
                ),
                kwargs=None,
                resources=Resources(),
                qualname="guess",
            )

        else:
            return ChoreSpec(
                args=(
                    guess_result["low"],
                    guess_result["guess"] - 1,
                    guess_result["num_guesses"] + 1,
                ),
                kwargs=None,
                resources=Resources(),
                qualname="guess",
            )

```

152 This mechanism allows MatEnsemble workflows to adapt their execution path based on
 153 intermediate results rather than requiring the entire workflow graph to be known before
 154 execution begins.

155 Research impact

156 AI usage disclosure

157 The package and wrapping API design are that of the author's and are the result of incremental
 158 development and testing. However, generative AI tools have been used during development:
 159 1) Microsoft Copilot integration in GitHub Pull Requests has been used for code reviews to
 160 ensure correctness and catch any extra bugs; 2) OpenAI's ChatGPT 5.4 & 5.5 has been used
 161 for "rubber duck debugging", general debugging, and for limited generation of boilerplate
 162 code, some tests and documentation. After the initial skeleton was established by the author's
 163 many methods were used to take advantage of the power of these models while ensuring

164 consistency. Methods such as “Spec Driven Development” and “Rubber Duck Debugging”. No
165 AI generated content has been accepted without review and testing, nor does it constitute the
166 majority of the work produced. For this paper, ChatGPT was used to suggest edits for clarity
167 and structure, but was not used to generate the contents or sections, nor were the suggestions
168 implemented wholesale.

169 **Acknowledgements**

170 **References**

171 [Flux Documentation](#) [JobFlow](#) [Parsl](#) [libEnsemble](#)

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